Alternative: Combinatorial Proof

We have already seen that set with n elements has a total of 2^n different subsets (*powerset*). Each subset has either zero elements, or one element, or two elements, …, or n elements in it.

There are $\binom{n}{0}$ subsets with zero elements, $\binom{n}{1}$ subsets with one element, $\binom{n}{2}$ subsets with two elements, …, and $\binom{n}{n}$ subsets with *n* elements.

Therefore,

$$\sum_{k=0}^{n} \binom{n}{k}$$

counts the total number of subsets of a set with n elements, which is 2^n.